

8-bit charge scaling DAC design report

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1. ABSTRACT

This report details the design process of an 8-bit charge scaling digital-to-analogue converter (DAC) which uses a split array topology with using gpdk 45nm technology. The DAC achieves a good conversion accuracy with low noise, low power and small device area.

2. INTRODUCTION

A DAC is a crucial component in modern electronic devices that converts digital data into analogue signals. It requires high precision and low noise operation to restore the analogue signal without degrading signal quality. And it typically consists of a **serial-to-parallel converter**, a **DAC** and a **buffer**. Being able to design a complete DAC with robust performance is an important step in becoming an electronic engineer.

The table below summarizes the performance specification of the DAC:

MaxV	MinV	P	DNL&INL	area	prop. delay
1V	0.1V	<5mW	<2LSB	< 5mm ²	< 1us

The available DAC architecture that best meets the design requirements are: **R2R DAC**, **current steering DAC** and **charge scaling DAC**.

3. SYSTEM OVERVIEW

TO assess the pros and cons of each possible DAC architectures, a thorough theoretical comparison along with simulations are performed to assist the initial decision making.

The R2R architecture can first be ruled out because: (1) Unlike current steering DAC, R2R requires an opamp, which can adds up to the already large power consumption of the DAC, making it unlikely to satisfy power requirements. (2) Unlike charge scaling DAC, increasing the size of components does not guarantee a better linearity. (3) Resistors occupies a similar amount of area as capacitors; even if one tries to limit power consumption by increasing resistor sizes, the area efficiency will decrease with a limited power compensation.

Current steering and charge scaling DAC offers similar advantages, and after further assessment: (1) current steering DAC doesn't use an opamp, making it prone to noise. (2) trying to improve matching and parasitics involves increasing the transistor sizes, making it equally area inefficient as the charge scaling DAC (3) power consumption is hard to estimate (4) The high frequency benefit is obsolete as the specification only requires a 400kHz sampling rate. (5) The specification requires DAC area to be $\leq 0.5\text{mm}^2$. This is a generous area

constrain, meaning the area issue for charge scaling DAC will not be a problem.

Based on these comparisons, the charge scaling DAC is used as it can achieve similar area efficiency and linearity as R2R and current scaling DAC. In terms of speed, it can achieve a similar performance as the R2R DAC. It also offers low-power and low noise operation. After considered those advantages, the trade-offs of large parasitics and limited bit resolution can be well compensated.

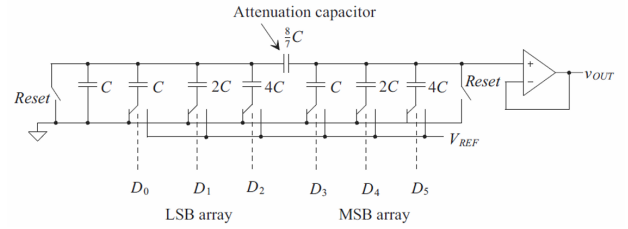


Fig. 3.1. Schematic design for the split array charge scaling DAC. The actual architecture used is 8-bit with LSB capacitor connected to $D_0 - D_3$ and MSB capacitors connected to $D_4 - D_7$. The value of the attenuation capacitor should be $\frac{C_{LSB}}{C_{MSB}^{sum}} C$. For the 8-bit case, the value is $\frac{16}{15} C$.

The attenuation capacitor in Figure 3.1 effectively scales down the LSB capacitances by $2^{\frac{N}{2}-1}$, so that the effective capacitances are: $\frac{C}{8}, \frac{C}{8}, \frac{C}{4}, \frac{C}{2}, C, 2C, 4C, 8C$ for the 8-bit case. Hence, the expression for the output voltage is still $\frac{C_{on}}{C_{total}}$. If all the capacitors are connected to Vref, the output is 255 and when all capacitors are connected to GND, the output should be 0. Furthermore, since the LSB capacitors have the same size as MSE capacitors, the improvement in area efficiency is:

$$\frac{\sum_{n=0}^{N-1} 2^n}{2(\sum_{n=0}^{\frac{N}{2}-1} 2^n) + 16} = \frac{1 + 2 + \dots + 128}{2(1 + 2 + 4 + 8) + 16} = 5.5$$

Similar to an R2R DAC, the output voltage equals to binary ratios of the capacitors, specifically the ratio between the total capacitance connected to V_{ref} and the total capacitance connected to GND :

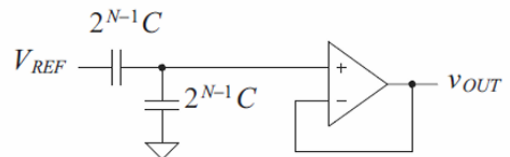


Fig. 3.2. Equivalent capacitance assuming MSB (D_{N-1}) = 1.

The general design strategy is to minimize the sizes of the repeating devices (e.g. capacitors) while maximizing the sizes of single devices (e.g. opamp) for optimal area utilization.

4. CIRCUIT IMPLEMENTATION

Switch design

The initial choice was to use a differential CMOS pair as the switch to select between V_{ref} and ground. This is the simplest design and offers maximum area efficiency. But there're two critical drawbacks: (1) The internal resistance of the transistors means that the switch can only be partially on or off, causing large offset and gain error. (2) The internal capacitance of the transistors makes switching really noisy, increasing DNL and INL errors.

Solution: Use a 2:1 multiplexer. Multiplexers can accurately select between either end of the input, making it a good choice for mimicking a switch. The first design attempt is to use a 2 transistor CMOS transmission gate, which achieves the same function as a 2:1 multiplexer. However, a 2 transistor design is still limited by low output swing. Therefore, a further improvement is to use a 2 sided transmission gate pair shown below. This enables the switch to be fully on or off with no limits in output swing. (See Appendix D.4 for more details)

During switching, capacitor transient voltages cause output voltage "spikes" due to current drawn from the DAC's output to maintain voltage continuity. (see Appendix C, Fig C.2) This can be mitigated by using larger transistors, balancing noise and area efficiency. Additionally, a unity-gain opamp buffer with high gain and low slew rate can smooth the output.

Design of unit capacitor

The most important component in the circuit which directly affects performance and area, it also has the following design trade-off:

- 1) Increasing the size of unit capacitor improves output range, but decreases area efficiency.
- 2) Reducing the unit capacitor sizes improves linearity but decreases output range

Issues encountered with unit capacitor design:

Using the split array topology, one can increase the size s of the unit capacitor to $15\mu m$ so the total area occupied by the capacitors is around $0.0133mm^2$. Note that the circuit performance stops improving for $s > 20\mu m$. However, the maximum size of mimcap for gpdK 45nm technology is limited to $5.6\mu m$. One solution is to use moscaps, however, these capacitors have larger process and mismatch variation and are difficult in terms of layout design. Hence the final solution is to build a custom unit capacitor but combining 8 or more mimcaps to form a larger capacitor. The custom unit capacitor can be used in the same fashion as a single mimcap. (see Appendix E for capacitor layout)

Attenuation capacitor

To achieve a $\frac{16}{15}C$ capacitance, a simple way is to connect 15 unit capacitor in series together with 1 unit capacitor in

parallel. (see Figure D.10)

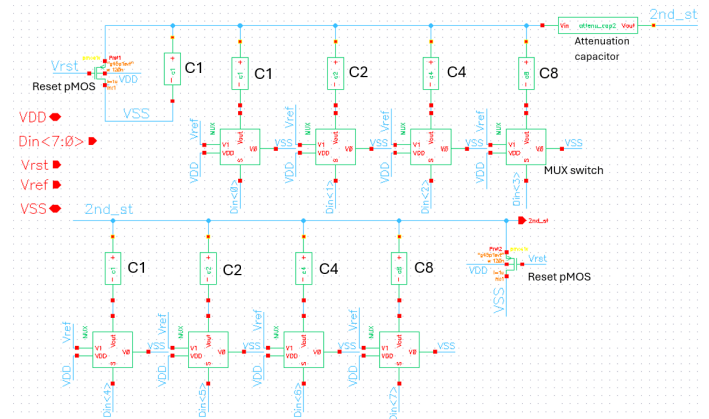


Fig. 4.1. The internal schematic of the charge scaling DAC.

Design of opamp buffer

The main function of the opamp is to provide a strong and stable feedback loop. The design of the opamp involves the following trade-offs:

- 1) The gain and phase margin must be high enough to stabilize the output but not too high, otherwise there may be oscillations and distortions in the output voltage.
- 2) GBP and slew rate can be compromised as only around 400kHz of bandwidth is needed.

However, following several experiments it was found that:

- (1) The phase margin should be between 45-55 degrees, otherwise there may be oscillations or jumps in the output.
 - (2) The gain should be at least 40dB but not more than 50dB.
- A performance table below summarizes the characteristics of the final opamp design:

DC gain	Phase margin	GBP	Slew rate
39.64dB	53.71°	1.581MHz	2.445V/us

See Appendix D for more details on opamp design.

Reset switch

The reset switch should ideally create an open circuit between the capacitors and gnd! when Rst = low, and shorts the capacitors to gnd! when Rst = high. In practice, a pMOS with large output resistance can be used to mimic the function of such a switch.

Serial-to-parallel (S2P) converter

The DAC receives a serial input of 8-bit data stream separated by 2.5us of delay. The serial data is converted into parallel bits to be fed into each capacitor simultaneously. The architecture used for S2P is the double shift register architecture. The idea is that the output are only updated once all the data have been aligned at the correct time position (the "data ready" time) (see Figure 5.2). The enable signal then samples at every "data ready" point to restore all the parallel bits of the serial data.

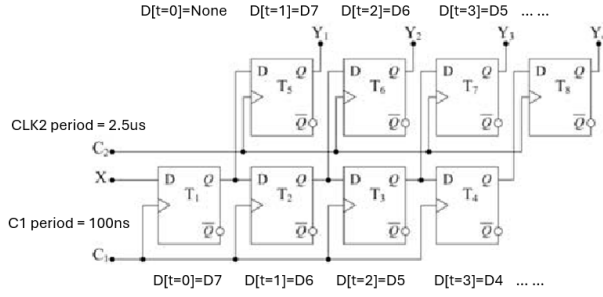


Fig. 4.2. The double shift register S2P is the simplest architecture with fastest conversion time and minimal output signal distortion. The downside it that it uses twice the amount of D-flip flops. This is compensated by that fact that all the components in the D-flip flop uses minimum size transistors to enable fast switching and minimise parasitics. For details of the D-flip flop schematics please refer to Appendix D.

Standard cells design

To maximize layout design efficiency, some parts of the circuit uses a hierarchical design with repeated standard cells to save time re-wiring identical components. Digital components such as the shift register and the switches are not very sensitive to mismatch, hence a symmetrical design is not essential.

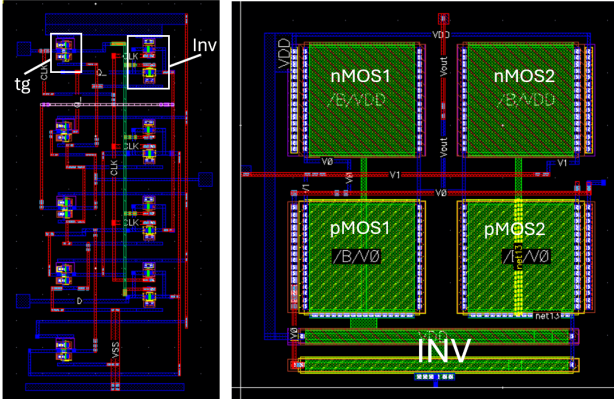


Fig. 4.3. Layout of D-flip flop (left) and the 2:1 multiplexer switch (right).

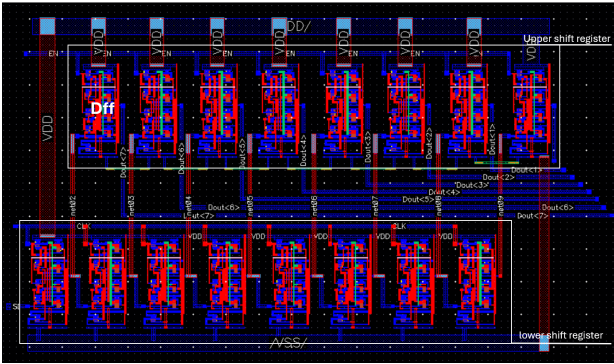


Fig. 4.4. Layout of the serial-to-parallel converter uses repeated connections of D-flip flops to simplify the design process.

The matching for the capacitors are more crucial. So we used a 16 x 16 grid layout to improve symmetry. The capacitors are laid out in a 1:2:4:8 ratio. The wiring uses a preferred

routing direction for easy wiring management. The power lines such as VDD or Ibias were made thicker.

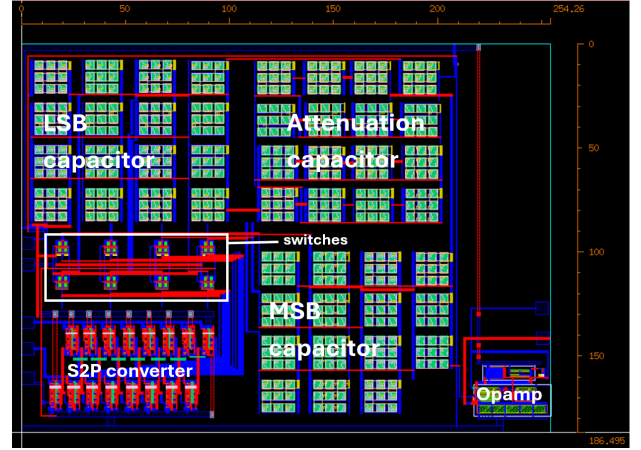


Fig. 4.5. Layout of full DAC circuit which measures $254.26\mu\text{m} \times 186.5\mu\text{m}$ in size and 0.04742mm^2 in area.

5. SIMULATED RESULTS

Linearity and basic function

We first verify the output quality of the DAC by building up a simple test bench that takes directly a parallel 8 bit data generated using a verilog block. This saves simulation time, as simulating with S2P converter may take hours.

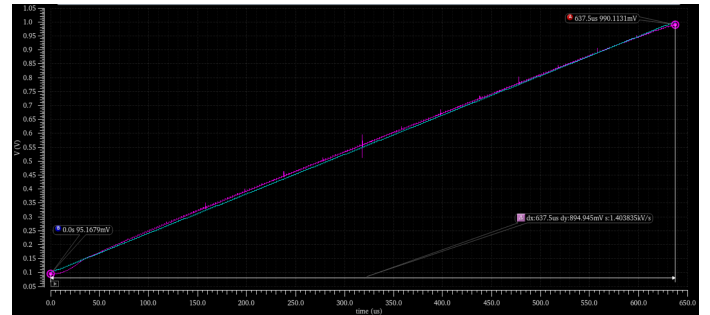


Fig. 5.1. A comparison between the DAC output and the ideal step output (DNL). Due to the time constant of the capacitors, there's a slight curve in the middle of the conversion time. To improve linearity, we can shift the output voltage curve down so that the curvature if divided by a half by the perfect output. The trade-off is that the output voltage headroom is reduced.

There's a large spike in the middle of the figure. This is because most capacitor switching occurs at $V_{out} = \frac{2^{N-1}}{2}$. Although the effect of spiking has been minimized by making the capacitors and switching transistors large (see appendix D). However, the trade-off is a slow response during the first 40us.

The table below summarizes the performance specification of the DAC:

MaxV	MinV	ave. P.	DNL	INL
990.1mV	95.17mV	165.3uW	6.92mV	14.29mV

Increasing the maximum output voltage swing reduces linearity and DNL/INL because the capacitors are charged to

a greater extend. Hence design parameters are chosen to not compromise maximum voltage too much while kept DNL to just below 2LSB (7mV). The settling time after applying a step input was found to be 4.67us. The propagation delay is 402.3ns for the DAC itself (see Appendix C.3 and C.4).

Calculating DNL and INL

DNL is the difference between the actual step output and the ideal step output; while INL is the difference between the actual step output and a reference line connecting the first and last points of the output. In this experiment a constant using the time averaged absolute difference between the actual and ideal outputs is calculated to measure a DNL and INL. Using the absolute value prevents the positive and negative values from cancelling each other out. The signed DNL and INL are included in Appendix C. The formula used to calculate DNL and INL for the previous experiments are shown below:

$$DNL = average(|V_{out_{actual}}(t) - V_{out_{ideal}}(t)|)$$

$$INL = average(|V_{out_{actual}}(t) - V_{out_{line}}(t)|)$$

For the INL calculation, since we have obtained the minimum and maximum of the output voltage as 990.1mV and 95.17mV, the reference line for INL is a straight line starting from (0us, 95.17mV) and ends at (637.5us, 990.1mV). This is easily generated using a pulse voltage source with a rise time of 637.5us ($255 \times 2.5us$). For detailed plots of DNL and INL see Appendix C.

Simulation with S2P

Simulate DAC with S2P with 0000 0001 resolution will take too much time. So we used a coarse step function to assess the conversion characteristics of the DAC which makes it easier to make bit-wise analysis. Hence the purpose of this simulation is less about linearity but more about verifying the performance of the DAC under S2P and large step changes. The experiment generates a serial input with increment of 0000 1111 (15).

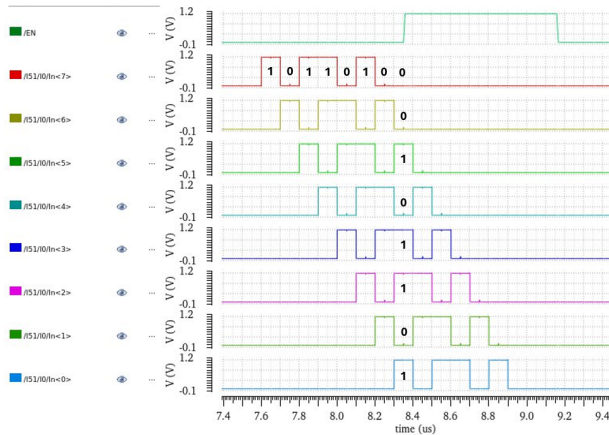


Fig. 5.2. The output of the bottom row shift register: $In < 7 : 0 >$. The serial input is shifted by 1 bit period (100 ns) for 8 times (corresponds to 10MHz clock speed). The rising clock edge of the enable signal at the second stage of the shift register should be located at the 8th shift to sample the aligned parallel 8 bits.

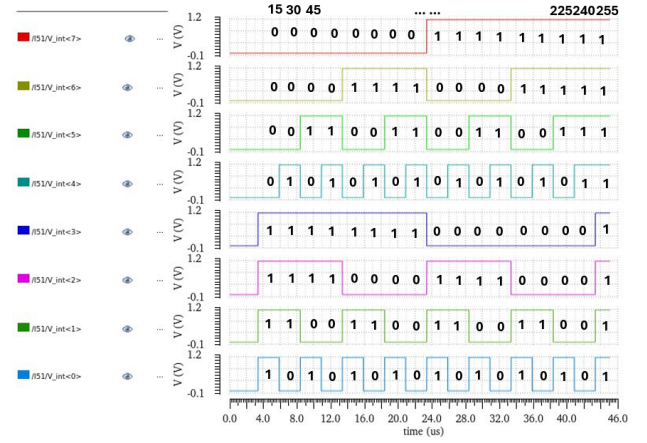


Fig. 5.3. Output of S2P converter: $V_{int} < 7 : 0 >$. This figure shows that the serial input is corrected converted into parallel bits. Step size 15 is chosen as it's also divisible by 255. For input serial bits see Appendix C.

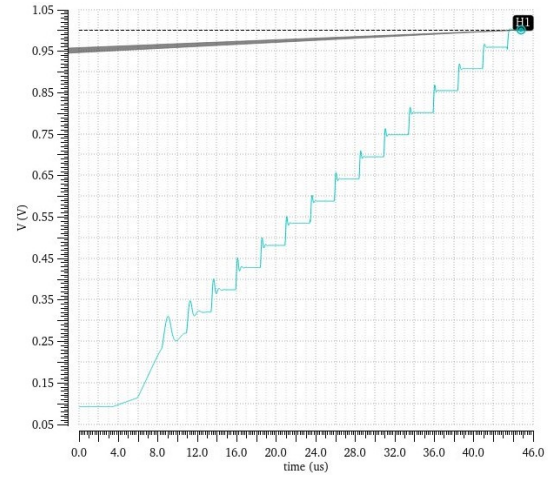


Fig. 5.4. The output step voltage with step size = 15 bits. The conversion accuracy at each level is high but takes time to stabilize due to the capacitor time constant. There's an initial delay in stepping as the capacitors takes time to charge up.

6. CONCLUSION

The DAC design meets most of the specifications while achieves an excellent power efficiency and are aefficiency. The design successfully mitigated the design issues in linearity, noise level control and layout matching. The DAC uses a custom serial-to-parallel converter along with near-perfect switches and a relatively compact device area. However, integral-nonlinearity can be improved by reducing the curvature of the output voltage. The settling time could be reduced by using smaller components. Also, there's a trade-off between suppressing oscillation and reducing the settling time, hence the charge scaling DAC is more suitable for continuous signals conversions.

APPENDIX A MONTE CARLO SIMULATION

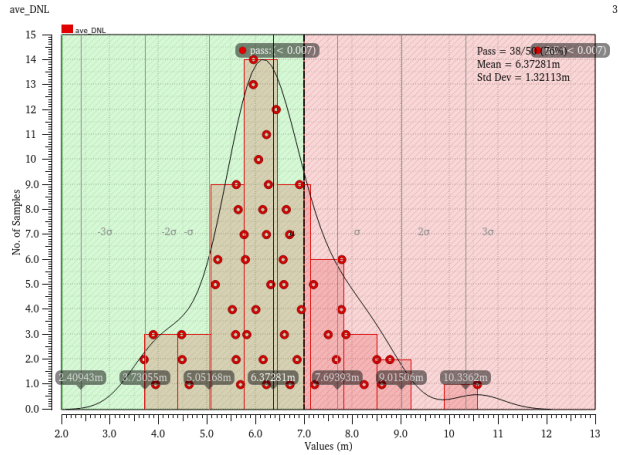


Fig. A.1. Monte Carlo - DNL

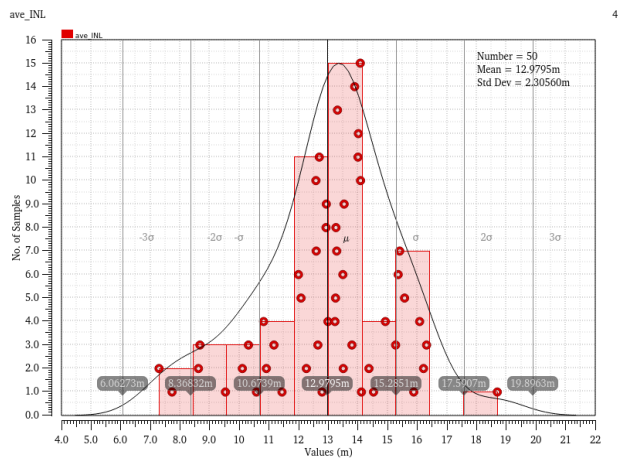


Fig. A.2. Monte Carlo - INL

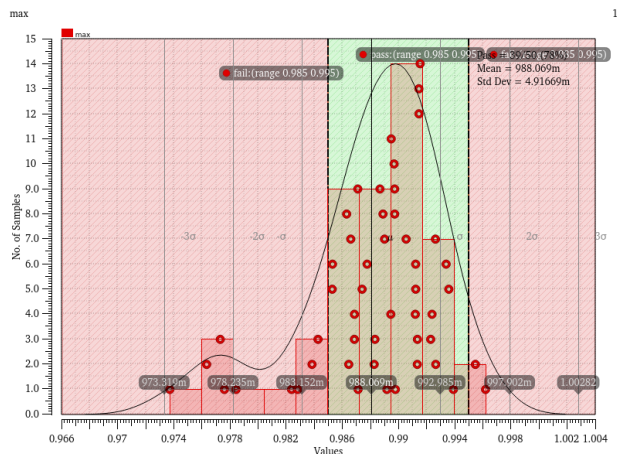


Fig. A.3. Monte Carlo - max. output voltage

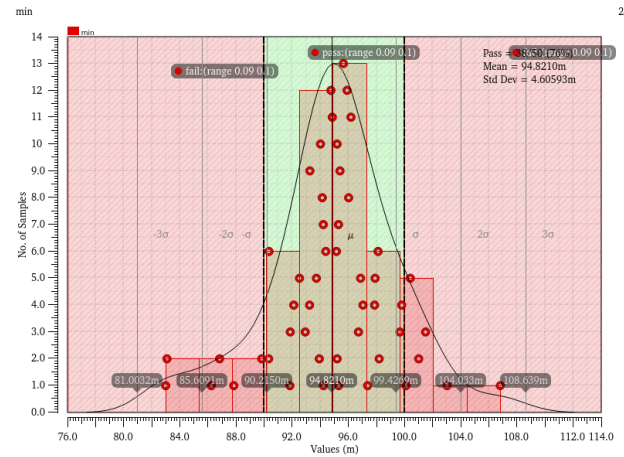


Fig. A.4. Monte Carlo - min. output voltage

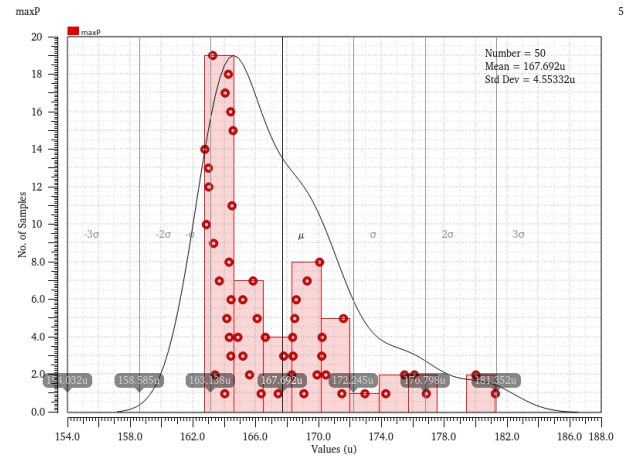


Fig. A.5. Monte Carlo - average power output

The table below summarizes the performance variation under mismatch and process Monte Carlo simulations.

	DNL	INL	maxV	minV	max P.
best	3.703mV	7.29mV	996.2mV	106.8mV	162.8uW
mean	6.37mV	12.98mV	988.1mV	94.82mV	167.7uW
std.	1.321mV	2.326mV	4.917mV	4.606mV	4.553uW

APPENDIX B CORNER SIMULATION

This section documented the DAC's performance under different transistor corners (fast/slow nMOS and fast/slow pMOS) and different temperature corners. In particular, this section assesses the circuit's linearity and output range under corner variations.

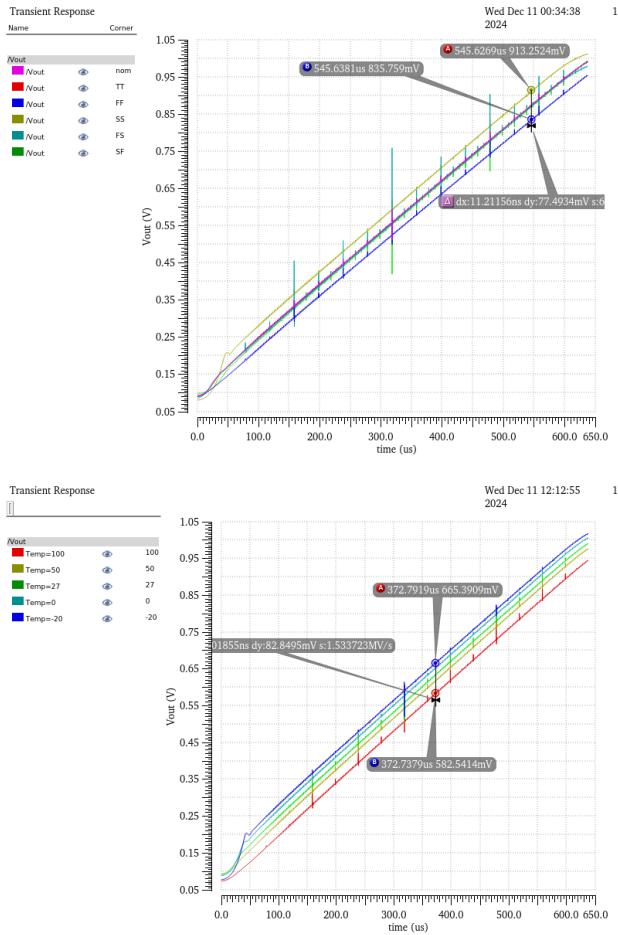


Fig. B.1. DAC's output plot with staircase 8-bit input under transistor (top) and temperature (bottom) corners. The deviation between the best and worst case scenario is highlighted in the plot.

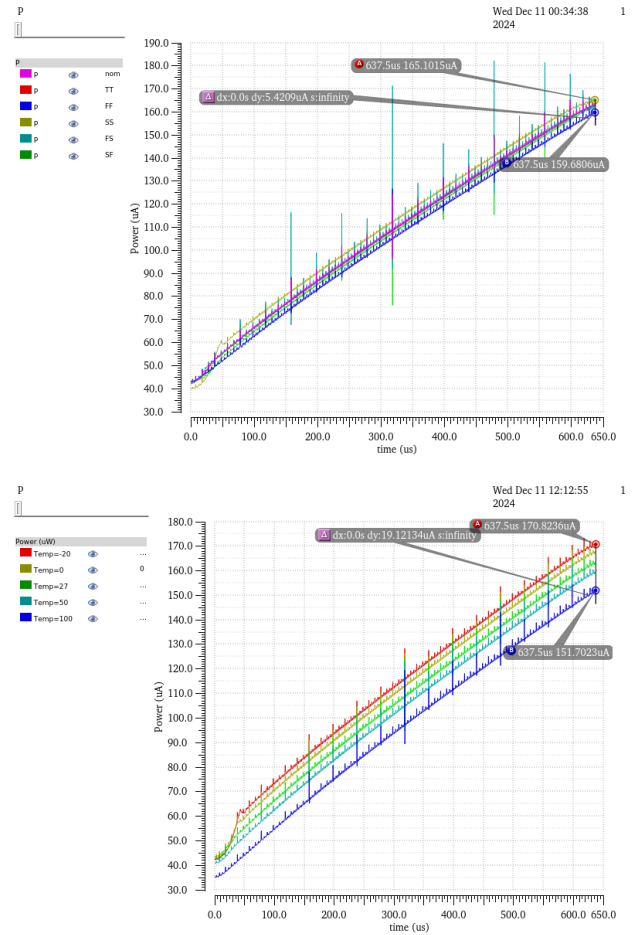


Fig. B.2. DAC's power output plot with staircase 8-bit input under transistor (top) and temperature (bottom) corners. The deviation between the best and worst maximum power is highlighted in the plot.

The 2 table below summarizes the complete performance variation under transistor and temperature corner simulations.

	DNL	INL	maxV	minV	max P.
TT	6.928mV	14.30mV	990.1mV	95.17mV	165.4uW
FF	27.56mV	18.91mV	953.6mV	92.32mV	162.9uW
FS	6.047mV	12.76mV	978.7mV	88.57mV	182.3uW
SF	4.19mV	10.34mV	992.9mV	101mV	163.4uW
SS	39.12mV	47.1mV	1012mV	83.32mV	167.8uW

Temp.	DNL	INL	maxV	minV	max P.
-20	36.77mV	44.81mV	1017mV	79.88mV	173.3uW
0	23.78mV	31.89mV	1006mV	91.14mV	169.9uW
27	6.928mV	14.3mV	990.1mV	95.17mV	165.4uW
50	10.53mV	3.103mV	976mV	91.84mV	162.1uW
100	44.32mV	35.67mV	944.3mV	76.23mV	154.5uW

Results that significantly deviates from the typical values are highlighted in bold.

APPENDIX C

ADDITIONAL SIMULATION RESULTS

This section displays some of the other performance characteristics of the DAC, as well as the simulation results of the opamp.

The 8-bit output for staircase simulation:

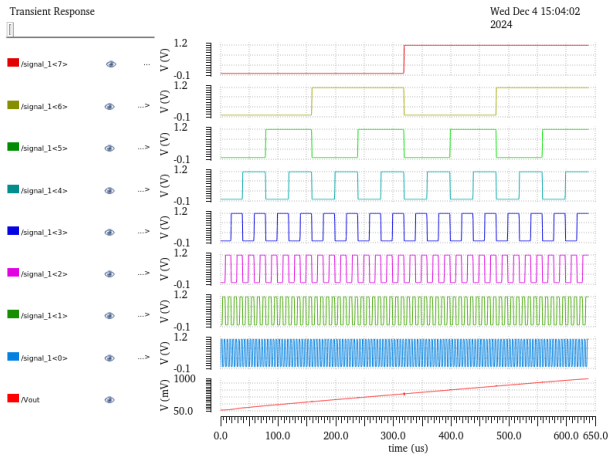


Fig. C.1. Output waveform of the staircase generator. Each bits generates a square wave pulse with frequency doubling for every lower bit. This effectively increments an 8-bit number from 0000 0000 to 1111 1111 with step size 1.

Dac's output voltage without opamp and load:

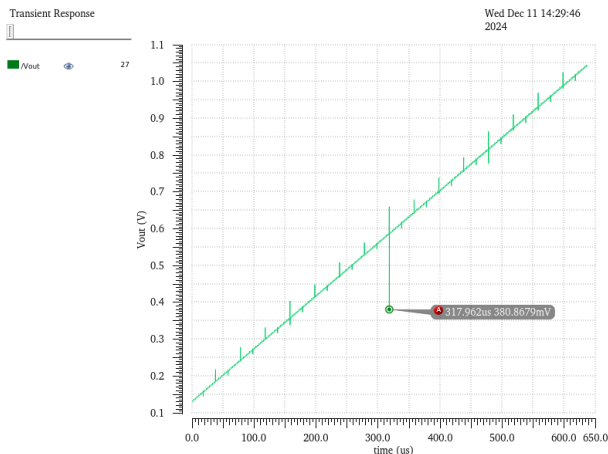


Fig. C.2. The linearity performance of the DAC without an unity buffer opamp or load. There's a large spike at the middle point of conversion time steps as that is when half of the capacitors switches simultaneously. There's also smaller spikes at time points where a significant number of capacitors switches on and off.

Settling time simulation:

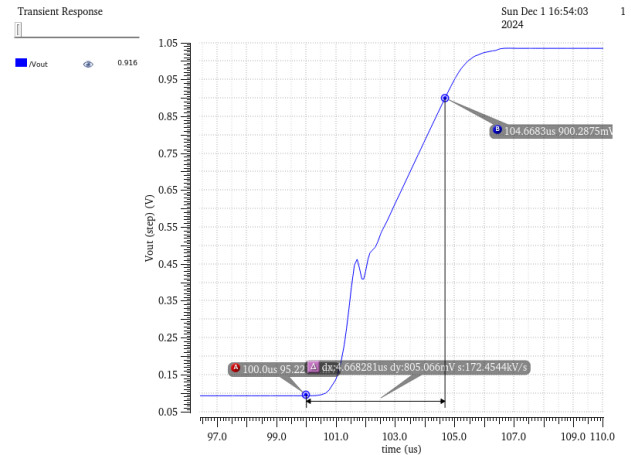


Fig. C.3. The step response of the DAC. Simulated by setting all the input bits to the DAC to high at once. This is to assess the DAC's settling time and subsequently the propagation delay.

Propagation delay simulation:

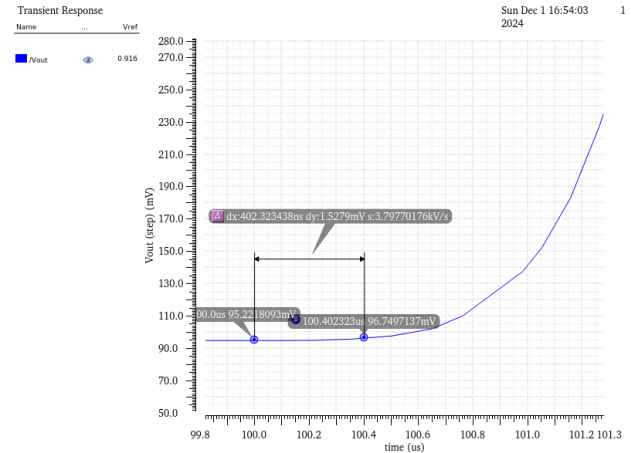


Fig. C.4. The amount of time taken for the DAC's output to respond after a step response. Note that this is the propagation delay of the DAC alone without opamp.

DNL and INL simulation:

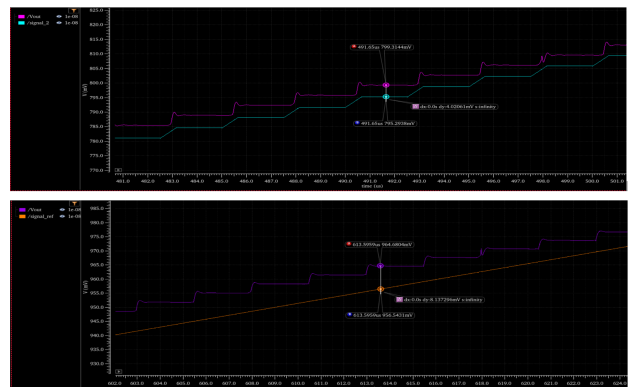


Fig. C.5. The zoomed in view of comparison actual output voltage with: (top) the ideal step output and (bottom) the reference line connecting the start and end point of the actual output voltage.

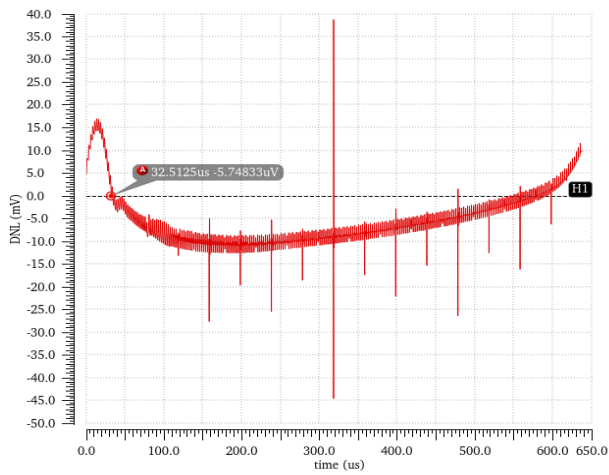


Fig. C.6. The Differential non-linearity plot.

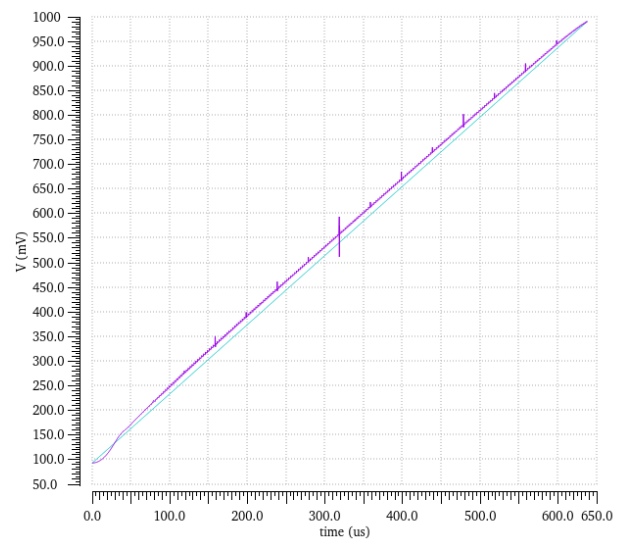


Fig. C.8. Macro-comparison between the actual output voltage (blue) and the reference line for INL (red).

Opamp simulation:

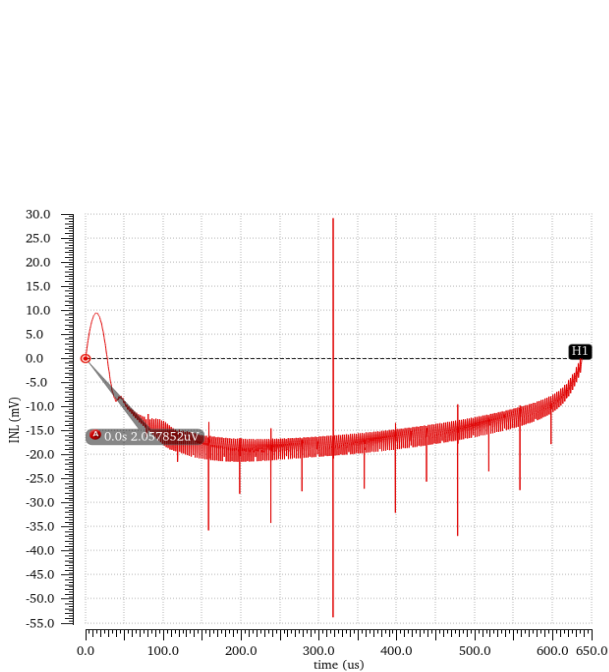


Fig. C.7. The Integral non-linearity plot.

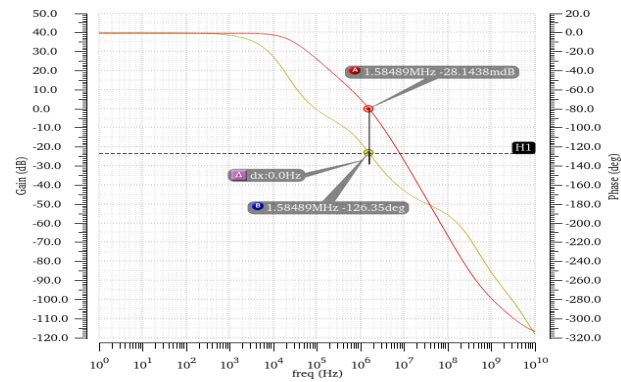


Fig. C.9. The Gain and phase response of the opamp. Simulated using open-loop configuration to verify the opamp's fundamental property and performance.

Serial input data simulation:

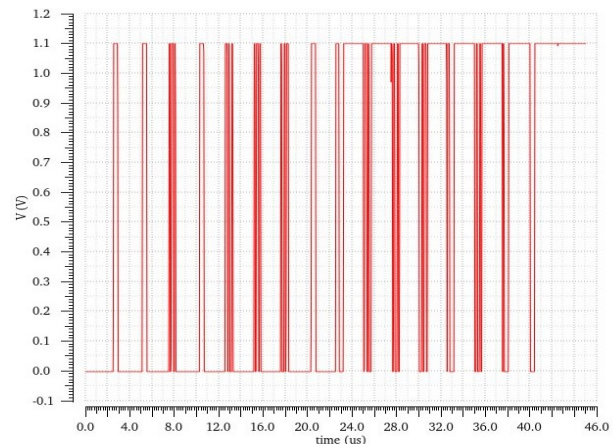


Fig. C.10. This plot shows the serial input data to the DAC. Generated using VerilogA code.

In general, INL is larger than DNL since the DAC's output voltage is curved upwards. This is likely due to the switching delay of the capacitors adding up, and this effect has been amplified further by the magnitude response of the opamp.

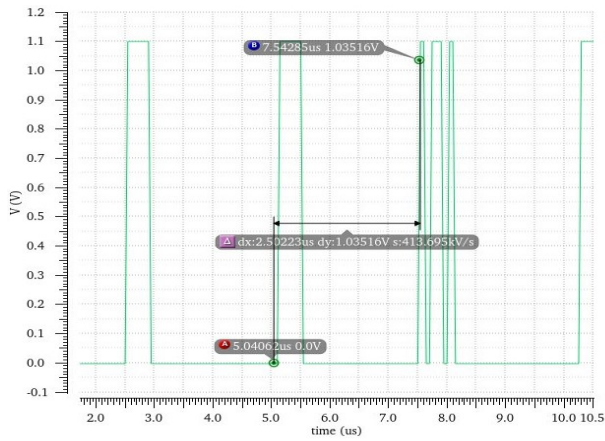


Fig. C.11. This plot shows a close up view of the serial input. Along with the correct time period of 2.5us (or a frequency of 400kHz).

Reset signal

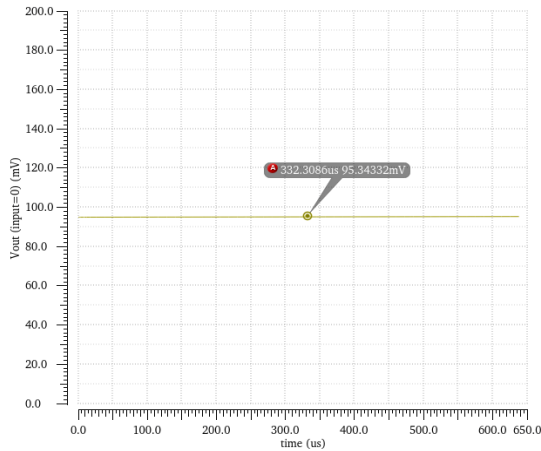


Fig. C.12. DAC's output when the input is 0V. A step input is applied to the 2 reset pMOS transistors at the start of the operation. The signal initially starts low which conducts the pMOS and discharges the capacitors. And after a short period, the signal goes high and isolate the capacitors from ground. The figure shows that when there's no input to the capacitors the reset switch sets Vout to the minimum voltage (around 95mV)

APPENDIX D INTERNAL SCHEMATIC SECTIONS

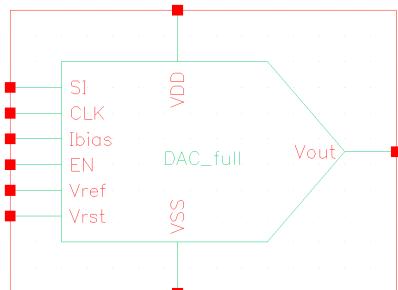


Fig. D.1. DAC symbol

The 3 main building blocks of the DAC:

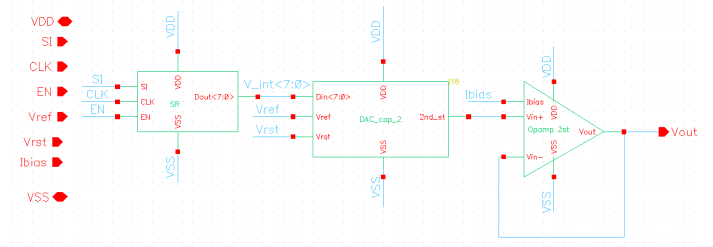


Fig. D.2. DAC subblocks with S2P converter, charge scaling DAC and output opamp buffer

Operational amplifier design:

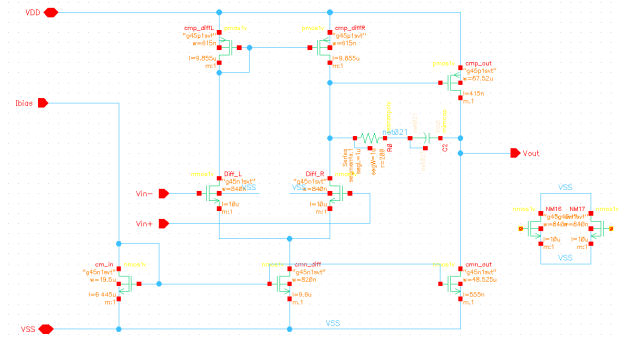


Fig. D.3. The schematic of the 2 stage opamp. Though this opamp uses an external bias current source. Future improvements could include replacing the ibias terminal with internally biased current source using a resistor connected via VDD. The width of the output pMOS is made as large as possible to maximize current gain. The width of the output nMOS current mirror, *cmn_out*, is made large to increase current gain. The Miller's compensation capacitor and a nulling resistor is used to improve phase margin and stability. The lengths of the differential pair and the pMOS current mirror are made much larger than its width to maximize input resistance. This is because the differential stage gain, $A_{v1} = -R_o g_{m_{Diff_R}}$ where $R_o = \frac{1}{g_o \frac{D_{iff_R}}{D_{iff_R}} + g_o \frac{C_{mp_diff_R}}{C_{mp_diff_R}}}$.

CMOS transmission gate

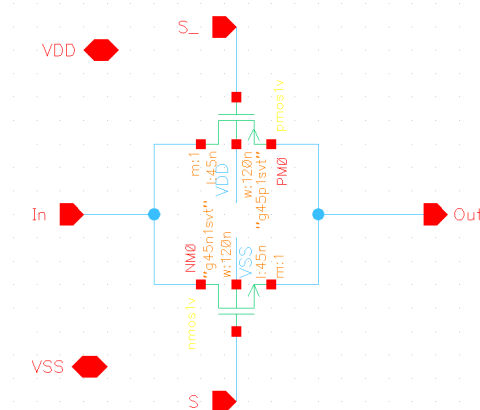


Fig. D.4. Transmission gate schematic. If S_- is high and S is low, pin In is connected to pin Out; if S_- is low and S is high, the signal path is blocked.

Switch design:

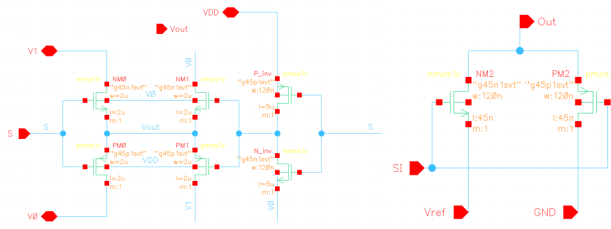


Fig. D.5. Schematic of dual transistor pair 2:1 multiplexer (left) and a differential CMOS switch (right). The differential switch has limited output swing and cannot fully switch off due to the internal resistance. The multiplexer uses 2 MOSFETs to improve voltage swing and achieves a better isolation ability.

D-flip flop:

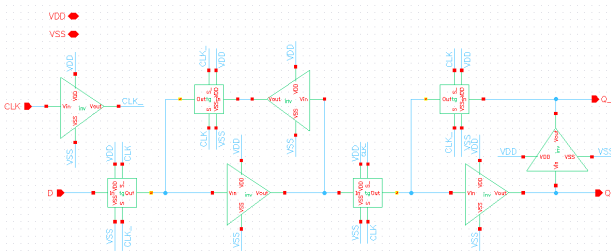


Fig. D.6. Schematic of the D-flip flop. When CLK is high, the signal propagate through the circuit via the positively connected transmission gates; when CLK is low, the feedback inverters and the negatively connected transmission gates creates a stable loop which stores the value just before CLK went low.

D-flip flop inverter:

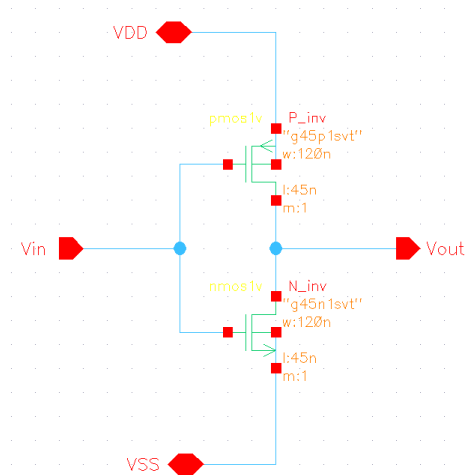


Fig. D.7. The inverter used in the D-flip flop differs from the ones used in the multiplexer as it uses minimum sizes to maximize switching speed.

Serial-to-parallel converter:

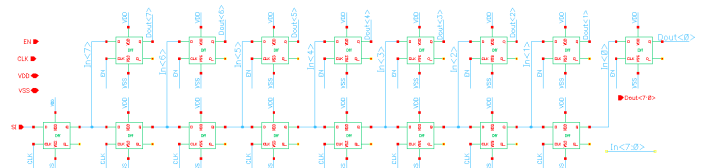


Fig. D.8. The schematic of the serial-to-parallel converter using 2 rows of 8-bit shift register. The 8-bit signal $In < 7 : 0 >$ is the intermediate signal after the first shift register and is used to verify the correctness of the conversion.

Testbench for simulating settling time response:

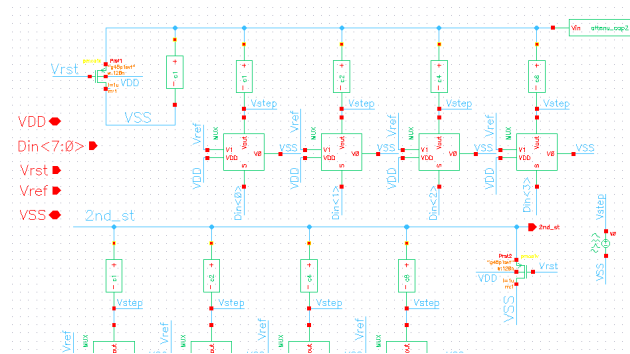


Fig. D.9. Test bench for evaluating settling time. Each capacitor is connected to the same step input to mimic the effect of all 8 bits going high at once.

Attenuation capacitor of charge scaling DAC:



Fig. D.10. Schematic of the attenuation capacitor.

The complete schematic view of the charge scaling DAC:

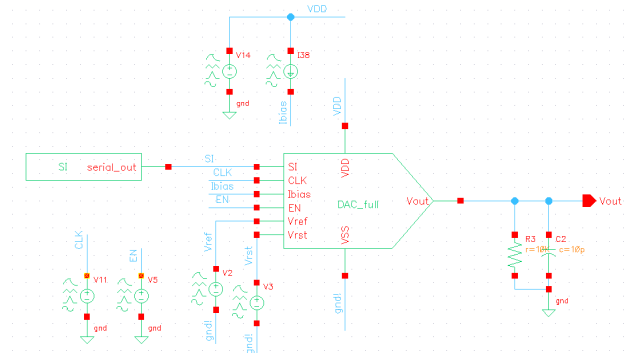


Fig. D.11. Schematics of DAC test bench with S2P, DAC and opamp.

Staircase simulation testbench for evaluation of linearity and conversion accuracy:

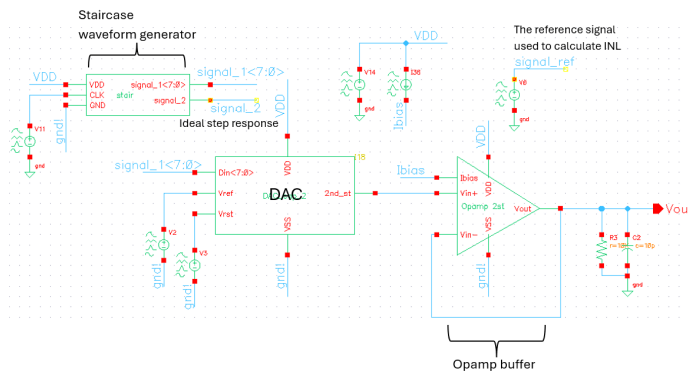


Fig. D.12. Schematics of DAC test bench with ideal staircase input, DAC and opamp. This is used in the *DAC_fast_tb* testbench file. The CLK of the staircase waveform generator is used set the sampling rate to 400kHz. If the amplitude of the CLK is set to 0, then the *signal_1* < 7 : 0 > input to the DAC will all be 0. This is used when we want to simulate the step response of the DAC or to assess how well the rst switch discharges the capacitors.

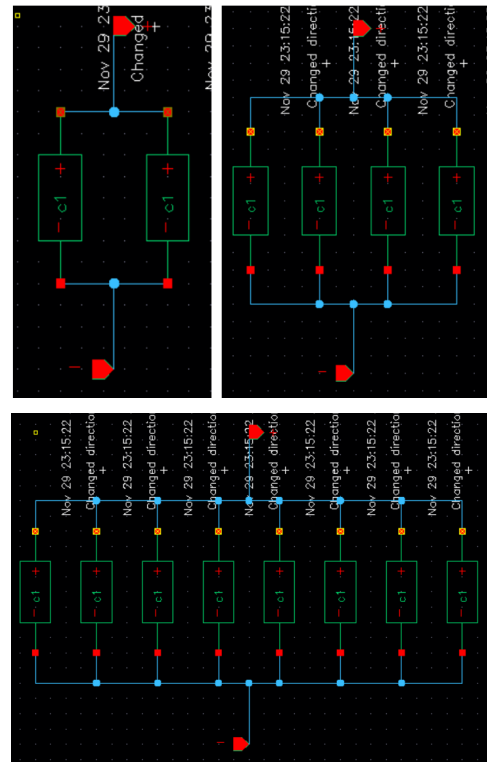


Fig. D.14. The unit capacitor is then connected in parallel to form larger capacitors. Future improvements can include using nMOS or pMOS capacitors to reduce area and saving the need to pre-build a compound capacitor.

The capacitors of charge scaling DAC:

APPENDIX E

INTERNAL LAYOUT SECTIONS

Unit capacitor: After extensive testing, the ideal capacitor dimension was found to be $\geq 12\mu\text{m} \times 12\mu\text{m}$, however the maximum sizes of a single mimcap is $5.6\mu\text{m} \times 5.6\mu\text{m}$. To address this limitation, 9 mimcaps with dimension $4\mu\text{m} \times 4\mu\text{m}$ are combined to form a larger $12\mu\text{m} \times 12\mu\text{m}$ capacitor.

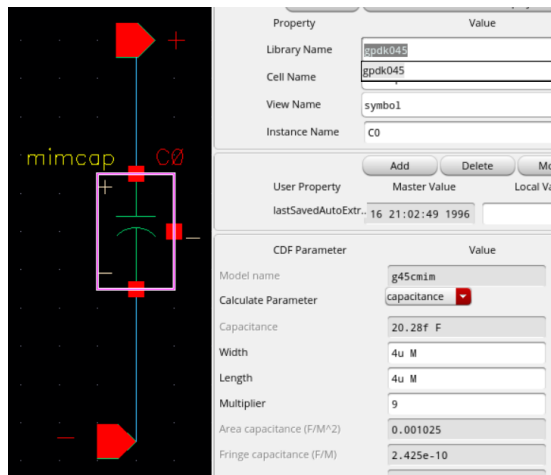


Fig. D.13. The unit capacitor is made of a multiple of nine $4 \times 4 \mu\text{m}$ mimcaps.

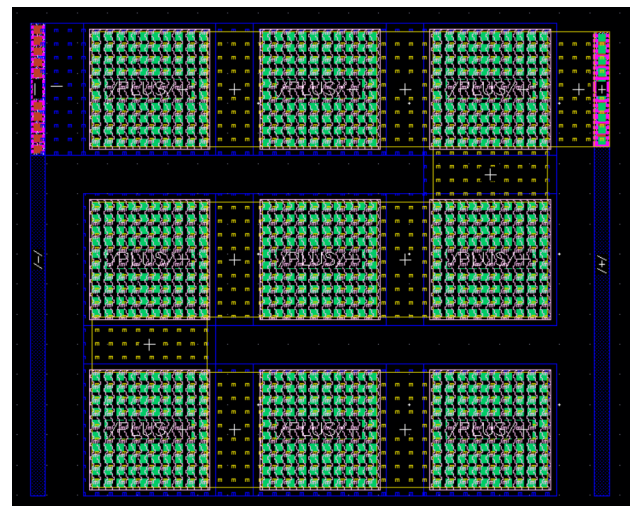


Fig. E.1. Layout of unit capacitor. The top and bottom metals of the 9 capacitors are connected together. This process is similar to designing a large metal resistor. The + and - terminals of the capacitor is kept unchanged.

Operational amplifier buffer:

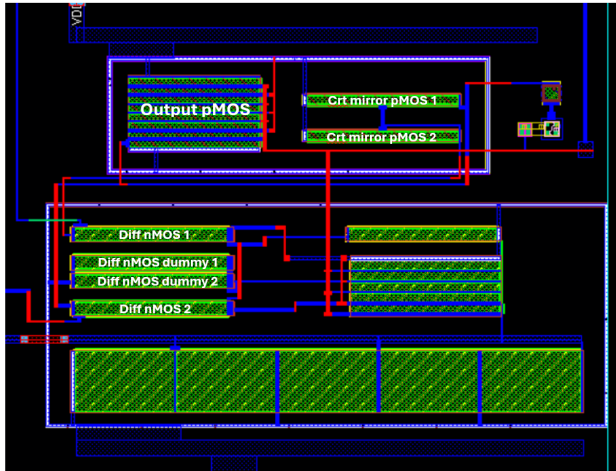


Fig. E.2. Layout section of the opamp. For the matching of the differential pair, instead of using split finger configuration, a pair of dummy transistor is used to enable an easier layout matching without having to re-wiring the split fingered devices. The pMOS current mirror pair does not require good matching since this is an unity buffer and does not need very high current.

Comelte DAC layout (named):

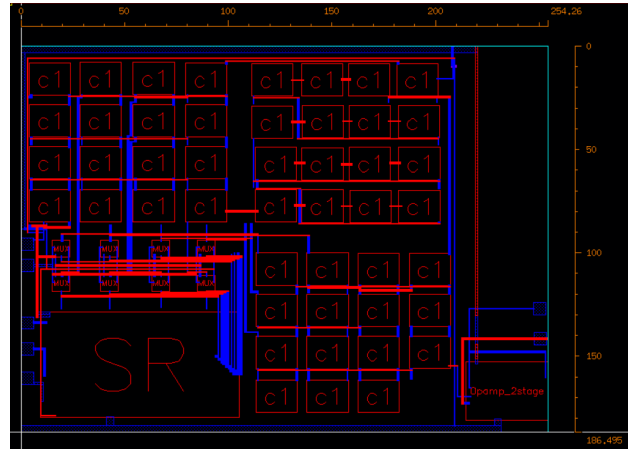


Fig. E.5. Layout with component names and their locations.

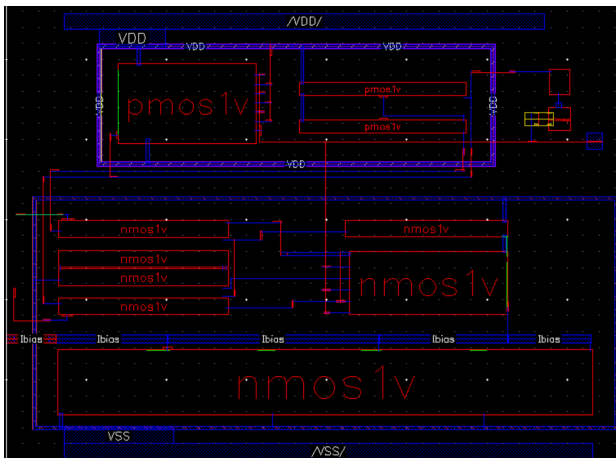


Fig. E.3. Layout of the opamp with sub-component names. From the diagram, the nMOS current mirrors all have different sizes so matching is not critical. The input nMOS which connects to the bias current, Ibias has the largest width to length ratio to generate a greater current gain.

Switches to the capacitors:

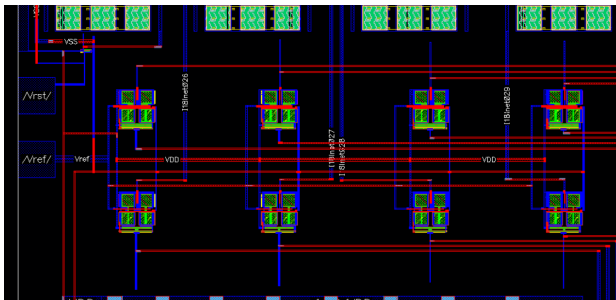


Fig. E.4. Layout section of how the multiplexer switches are connected.

APPENDIX F DRC & LVS

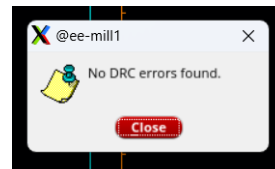


Fig. F.1. DRC results

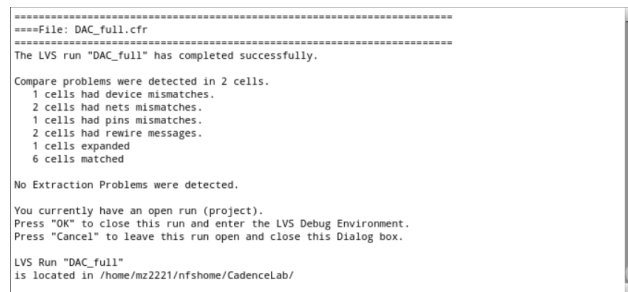


Fig. F.2. LVS results

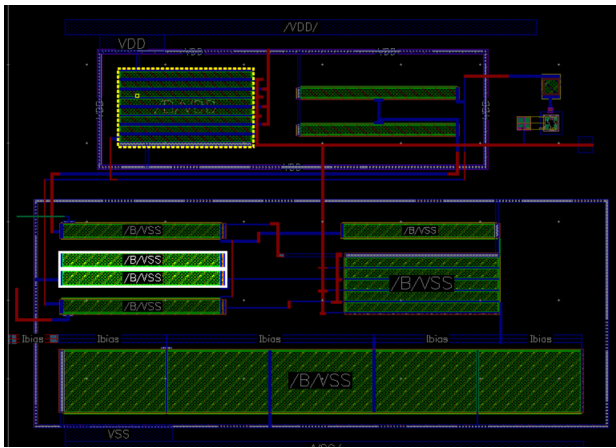


Fig. F.3. Sometimes Cadence does not recognize updated components/nets. The 2 device mismatch are caused by the 2 transistors being replaced by a new one.

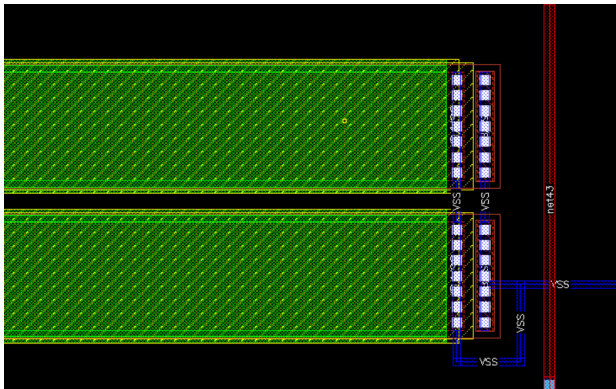


Fig. F.4. Hence even though the 2 transistors are correctly wired, LVS thinks that the nets are wrong as the old nets are not called VSS. The subsequent warnings are then caused by the net mismatch.

APPENDIX G LIBRARY MANAGEMENT

The schematic and layout of the opamp used for the DAC comes from a modification of the 2 stage opamp from Lab4 of the laboratory session. Hence there are 2 design libraries. The *backup* folder contains the old layout of the opamp for Lab4 only.

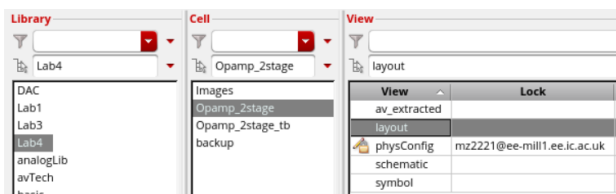


Fig. G.1. Library dependency.